Advanced Causal Inference

Lecture 03: Randomized Experiment

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This lecture note is partially based on the lecture notes/textbook of Hyunseung Kang, Ying Zhou, Ding (2024), Hernán and Robins (2020)

Outline

1. Observational Study vs Randomized Experiment
2. Randomized Experiment
3. Examples of Randomized Experiments
4. Model-based Inference: Infinite Population
5. Randomization-based Inference: Finite Population: Neymanian Inference
6. Randomization-based Inference: Finite Population: Fisherian Inference
7. Covariate Balance and Adjustment
8. Assignment

Observational study vs Randomized experiment

- Observational study: researchers simply observe or question the subjects
- Randomized experiments: researchers randomly assign the values of an action A to the subjects, and observe the effects of the action on a response variable Y

Example: Attitudes towards sentence

- The `attitudes` data frame in `R` package `tigerstats` contains the results of a large-scale survey conducted at Georgetown College in the fall semester of 2001

Questionaire

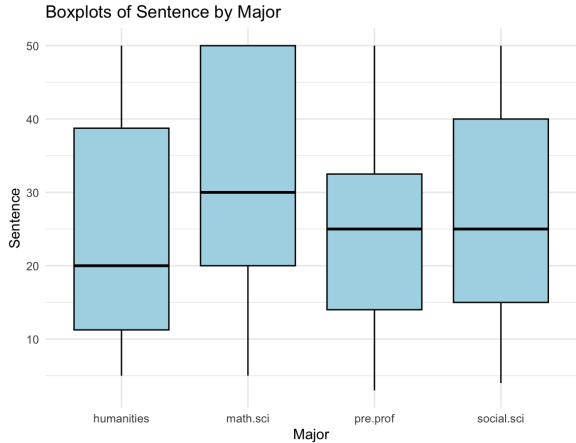
You are on a jury for a manslaughter case in Lewistown, PA. The defendant has been found guilty, and in Pennsylvania it is part of the job of the jury to recommend a sentence to the judge. The facts of the case are as follows.

The defendant, Tyrone Marcus Watson, a 35-year old native of Lewistown, was driving under the influence of alcohol on the evening of Tuesday July 17, 2001. At approximately 11:00 PM Watson drove through a red light, striking a pedestrian, Betsy Brockenheimer, a 20-year old resident of Lewistown. Brockenheimer was taken unconscious to the hospital and died of her injuries about one hour later. Watson did not flee the scene, nor did he resist arrest.

The prior police record for Mr. Watson is as follows: two minor traffic violations, and one previous arrest, five years ago, for DUI. No one was hurt in that incident.

Watson has now been convicted of DUI and manslaughter. The minimum jail term for this combination of offenses is two years; the maximum term is fifty years. In the blank below, write a number from 2 to 50 as your recommended length of sentence for Tyrone Marcus Watson.

Example: Attitudes towards sentence



- **Research Question:** Does studying in different major have a causal effect on hardness on crime?
- This is an observational study

Example: Attitudes towards sentence

- Respondents randomly assigned one of the four forms:

Defendant	Victim
Tyrone Marcus Watson	Betsy Brockenheimer
William Shane Winchester	Betsy Brockenheimer
Tyrone Marcus Watson	Latisha Dawes
William Shane Winchester	Latisha Dawes

- Tyrone and Latisha are often perceived as Black names, whereas William and Betsy are not
- **Research Question:** Does the suggested race of the defendant/victim have a causal effect on the length of sentence a subject would recommend?
- This is a randomized experiment

Other Examples

- Researchers test whether zinc lozenges help people with colds recover more quickly than taking no lozenges at all. They gather 40 people suffering from a cold and randomly assign 20 of them to take zinc lozenges. The remaining 20 take a lozenge made of an inert substance known to have no effect on the body.
  - A: zinc lozenges versus inert substance
  - Y: recovery time from cold

- Researchers gather a number of people and find that, of those who wear bifocals, 5% have colon cancer, whereas only 1% of those who do not wear bifocals have colon cancer.
  - A: wearing bifocals versus not
  - Y: colon cancer status

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Randomization Is the Gold Standard

- Since $\mathrm{E}\{Y(a)\} \neq \mathrm{E}(Y \mid A = a)$, we have $\tau_{\mathsf{ATE}} \neq \mathrm{E}(Y|A = 1) - \mathrm{E}(Y|A = 0)$

- Note that this discrepancy arises from the conceptual difference in the expectations:
  - $\mathrm{E}\{Y(a)\}$ is taken over the entire population;
  - $\mathrm{E}(Y|A = a)$ is taken over the specific subgroup with $A = a$

- Therefore, if the subgroup $A = a$ is the same as the population, we can say $\mathrm{E}\{Y(a)\} = \mathrm{E}(Y|A = a)$.
  How can we achieve this?

- **Randomization**: A is randomly assigned (i.e., independent of all other variables)
  - In particular, we have $A \perp\!\!\!\perp \{Y(1), Y(0)\}$
  - In addition, we have $A \perp\!\!\!\perp$ {unit's any measured/unmeasured characteristics}

- We have $\tau_{\mathsf{ATE}} = \mathrm{E}(Y|A = 1) - \mathrm{E}(Y|A = 0)$, because

$$\begin{aligned} \mathrm{E}\{Y(a)\} &= \mathrm{E}\{Y(a)|A = a\} && A \perp\!\!\!\perp Y(a) \\ &= \mathrm{E}(Y|A = a) && \text{SUTVA} \end{aligned}$$

Designs for Randomized Experiments

- Bernoulli trials
- Completely randomized experiments
- Stratified randomized experiments

Bernoulli Trials

- For each unit, $A_i \overset{iid}{\sim} \text{Ber}(p)$, where $p \in (0, 1)$
- 2^N possible choices for $(A_1, \ldots, A_N)$, and each of the 2^N possible choices has a known non-zero probability
- Limitations:
  - Assignments where all units receive the same treatment have positive probability, i.e., $\Pr(A_1 = \cdots = A_N = 1) > 0$
  - The number of treated and untreated units can be imbalanced by chance, even if $p = 0.5$

Completely Randomized Experiment

- Exactly N_1 of N units receive treatment and the remaining $N_0 = N - N_1$ units receive control, with each assignment equally likely

- The treatment vector $\boldsymbol{A}$ is uniform over the set $\{\boldsymbol{a}^1, \ldots, \boldsymbol{a}^M\}$
  where $M = \binom{N}{N_1}$, and $\boldsymbol{a}^m$ are all possible vectors with N_1 1's and $N - N_1$ 0's with $\Pr(\boldsymbol{A} = \boldsymbol{a}^m) = 1/\binom{N}{N_1}$

  - e.g., $N = 5, N_1 = 3$, $M = \binom{5}{3} = 10$

```
> permutation10(5, 3)
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
[1,]    1    1    1    1    1    1    0    0    0     0
[2,]    1    1    1    0    0    0    1    1    1     0
[3,]    1    0    0    1    1    0    1    1    0     1
[4,]    0    1    0    1    0    1    1    0    1     1
[5,]    0    0    1    0    1    1    0    1    1     1
```

- This eliminates "unhelpful" assignment vectors that are unlikely to yield precise causal inference.

- How does this compared to Bernoulli trials?

CRE vs Bernoulli trials

- Conditional on (N_1, N_0) in a Bernoulli randomized experiment makes it equivalent to a completely randomized experiment.
- **Completely Randomized Experiment:** The probability of any assignment vector with $\sum_{i=1}^{N} a_i = N_1$ is

$$\Pr(\boldsymbol{A} = \boldsymbol{a}) = \frac{1}{\binom{N}{N_1}}$$

- **Bernoulli trials:** The probability of any assignment vector with $\sum_{i=1}^{N} a_i = N_1$ is

$$\Pr\left(\boldsymbol{A} = \boldsymbol{a}, \sum_{i=1}^{N} a_i = N_1\right) = p^{N_1}(1-p)^{N_0}$$

The conditional probability of $\boldsymbol{A} = \boldsymbol{a}$ given $\sum_{i=1}^{N} a_i = N_1$ is

$$P\left(\boldsymbol{A} = \boldsymbol{a} \,\middle|\, \sum_{i=1}^{N} a_i = N_1\right) = \frac{p^{N_1}(1-p)^{N_0}}{\binom{N}{N_1} p^{N_1}(1-p)^{N_0}} = \frac{1}{\binom{N}{N_1}}$$

Completely Randomized Experiment

- Limitations:
  - Consider a study with $N = 20$ units: ten men and ten women.
  - The potential outcomes $Y(1)$ and $Y(0)$ are expected *a priori* to vary substantially by sex.
  - We want to estimate causal effects for each sex
  - Even with a completely randomized design assigning $N_1 = 10$ units to treatment, it is possible that all ten treated units are men (or all women), leading to imbalance

Stratified Randomized Experiments

- The population of units in the study is first partitioned into blocks or strata so that the units within each block are similar w.r.t. some (functions of) covariates thought to be predictive of potential outcomes

- Within each block, we conduct a Bernoulli trial/completely randomized experiment, with assignments independent across blocks

- Advantage:
  - When the stratification variable is predictive of the potential outcomes $(Y_i(1), Y_i(0))$, then randomizing within the strata will lead to more precise inferences
  - If the stratification variable perfectly predicts both potential outcomes $(Y_i(1), Y_i(0))$, then there will be no uncertainty remaining regarding the treatment effect across the N units or within the subgroups defined by the covariate
  - Even if there is no predictive power of the stratification variable, stratification does not reduce precision

- George Box: *"Block what you can; randomize what you cannot."*

Paired Randomized Experiments

- An extreme version of the stratified randomized experiment: each block contains exactly two units
- A fair coin is tossed to determine which member of the pair receives the active treatment and which receives the control treatment
- Example: York (2019)
  - Goal: To study how reading KKK-friendly $(A = 1)$ versus KKK-hostile $(A = 0)$ news articles affects individuals' opinions of the KKK
  - The experiment was conducted at the annual Twins Days Festival in Twinsburg, Ohio
  - In the original study, each twin pair was randomly assigned to $(A_1, A_2) = (0, 0), \ldots, (1, 1)$; alternatively, one could assign $(A_1, A_2) = (0, 1)$ or $(1, 0)$
- We will discuss this type of experiment later

Summary

- **Bernoulli trial/Conditional randomized experiments**
  - Each treatment A_i are independent of each other (within blocks)
- **Completely/Stratified randomized experiments**
  - Guarantee that N_1 of N units receive treatment
  - More efficient when the sample size N is small
  - The efficiency gain is negligible when the sample size is large

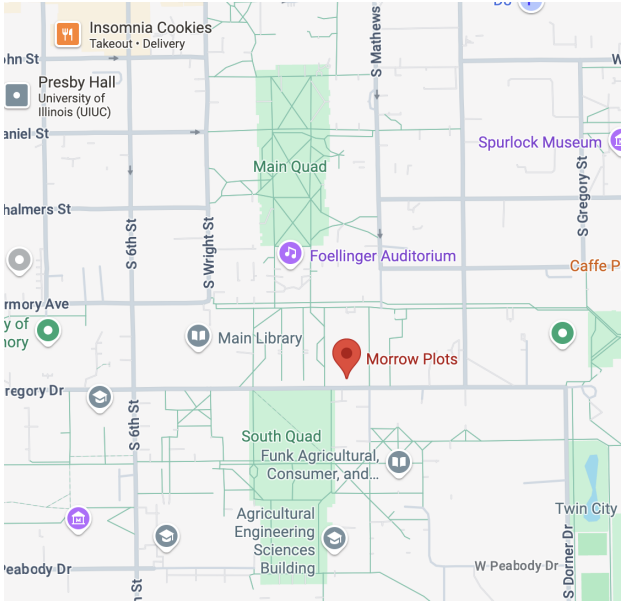
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Examples of Randomized Experiments: Morrow Plots

- Morrow plots

- Established in 1876, the Morrow Plots at the University of Illinois Urbana Champaign are the longest-running continuous agricultural fields in the Americas. They test the impact of soil fertility treatments and crop rotation on corn, Illinois's most important crop. Data collection began in 1888 and has recorded planting, treatment, and yield data annually since then.



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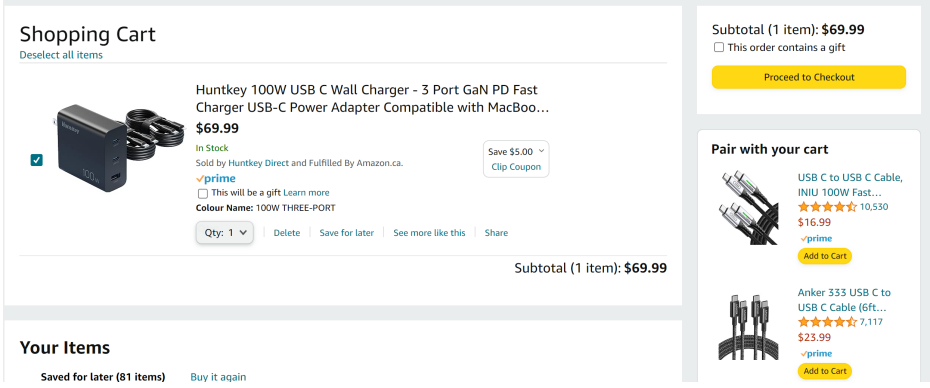
Examples of Randomized Experiments: A/B Testing

- A/B testing from Wikipedia:

  A/B tests consist of a randomized experiment that usually involves two variants (A and B), although the concept can be also extended to multiple variants of the same variable. It includes application of statistical hypothesis testing or "two-sample hypothesis testing" as used in the field of statistics. A/B testing is employed to compare multiple versions of a single variable, for example by testing a subject's response to variant A against variant B, and to determine which of the variants is more effective.

Amazon Shopping Cart Recs

- Add an item to your shopping cart at a website
- Most sites (at the time) show the cart
- At Amazon, Greg Linden (2006) had the idea of showing recommendations based on cart items evaluation



Amazon Shopping Cart Recs

- **Greg**: cross-sell more items (increase average basket size)

  HiPPO (Highest Paid Person's Opinion): distract people from checking out (reduce the conversion rate)

  - conversion rate: after seeing the website, the percentage of viewers who take the desired action (=buy products)
  - Greg was "forbidden to work on this any further"

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- He nevertheless prepared the feature for an online test
  - the sales vice president was angry
  - But, even for top executives, it was hard to block a test. The only good argument against testing would be that the negative impact might be so severe that Amazon couldn't afford it, a difficult claim to make. The test rolled out.

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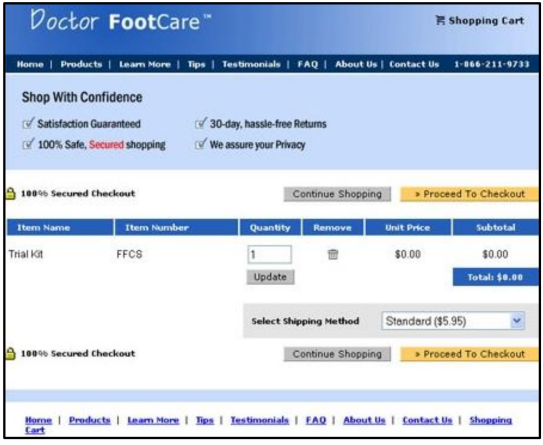
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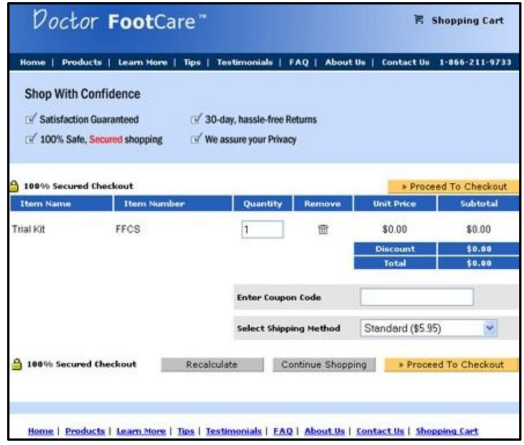
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- **the feature won by a wide margin**
  - With new urgency, shopping cart recommendations launched

Checkout page at Doctor FootCare

- Conversion rate: the percentage of visits to the website that include a purchase





- Which do you think result in a higher conversion rate?

Checkout page at Doctor FootCare

- The site “upgraded” their webpage from the left to the right
- Lost 90% of their revenue!

Checkout page at Doctor FootCare

- The site "upgraded" their webpage from the left to the right
- Lost 90% of their revenue!
- Most of the changes in the upgrade were positive, but the **coupon code** was the critical one:
  - people started to think twice about whether they were paying too much because there are discount coupons out there that they do not have
- Removing the discount code from the new version increases the conversion rate by 6.5 percent

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Recall: Two Sample t-Test

- $Y_{ij} \overset{iid}{\sim} (\mu_j, \sigma_j^2)$ for $i = 1, \ldots, N_j$ and $j = 0, 1$

- From the CLT, we have

$$\sqrt{\frac{N_j}{\widehat{\sigma}_j^2}}(\overline{Y}_j - \mu_j) \xrightarrow{D} N(0,1) \ , \qquad \overline{Y}_j = \frac{\sum_{i=1}^{N_j} Y_{ij}}{N_j} \ , \quad \widehat{\sigma}_j^2 = \frac{\sum_{i=1}^{N_j} (Y_{ij} - \overline{Y}_j)^2}{N_j}$$

- An estimator of $\mu_1 - \mu_0$: $\overline{Y}_1 - \overline{Y}_0$ with the following asymptotic distribution:

$$\frac{(\overline{Y}_1 - \overline{Y}_0) - (\mu_1 - \mu_0)}{\sqrt{\frac{\widehat{\sigma}_1^2}{N_1} + \frac{\widehat{\sigma}_0^2}{N_0}}} \xrightarrow{D} N(0,1) \ , \quad N_1, N_0 \to \infty \text{ with } N_1 \sim N_0$$

Model-based inference

- $Y_i(a) \overset{iid}{\sim} (\mu_a, \sigma_a^2)$ for $i = 1, \ldots, N$ and $a = 0, 1$
- Further assumes $A_i \overset{iid}{\sim} \text{Ber}(p)$
- Under randomized experiments, we have $\mu_a = \mathrm{E}\{Y(a)\} = \mathrm{E}(Y|A = a)$
  - Note: distinguish $=$ and $=$
- We know that a good estimator for $\mathrm{E}(Y|A = a)$ is the sample mean of Y among units with $A = a$
- From the CLT, we have

$$\sqrt{\frac{N_a}{\widehat{\sigma}_a^2}}(\overline{Y}_a - \mu_a) \xrightarrow{D} N(0,1) \ , \qquad \overline{Y}_a = \frac{\sum_{i:A_i=a} Y_i}{N_a} \ , \quad \widehat{\sigma}_a^2 = \frac{\sum_{i:A_i=a}(Y_i - \overline{Y}_a)^2}{N_a} \ , \quad N_a = \sum_{i=1}^{N} \mathbb{1}(A_i = a)$$

- An estimator of the ATE $\tau_{\text{ATE}} = \mu_1 - \mu_0$: $\widehat{\tau}_{\text{ATE}} = \overline{Y}_1 - \overline{Y}_0$ with the following asymptotic distribution:

$$\frac{\widehat{\tau}_{\text{ATE}} - \tau_{\text{ATE}}}{\sqrt{\dfrac{\widehat{\sigma}_1^2}{N_1} + \dfrac{\widehat{\sigma}_0^2}{N_0}}} \xrightarrow{D} N(0,1) \ , \quad N \to \infty$$

- Perform hypothesis testing and construct confidence intervals (omitted)

Model-based inference

- Observed data $(A_i, Y_i) \overset{iid}{\sim} P$
  i.i.d. based on a random sample from an infinite population P
- Assignment mechanism: Bernoulli trials
- Infinite population: **external validity**; causal conclusions can be generalized beyond the N study sample
- Commonly used mode of inference in practice
- Extends more easily to observational studies

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Randomization-based Inference

- Minimal assumptions: SUTVA, completely randomized experiment (CRE)
  - exactly N_1 of N units receive treatment, with each assignment equally likely
- No assumptions about the population / existence of a super-population
- Internal validity: the statistical analysis is valid for the study samples at hand

Randomness in CRE

- In randomization-based inference, $Y_i(1), Y_i(0)$ are treated as fixed quantities for $\forall i$
  - This can seem awkward at first, especially for those familiar with the i.i.d. framework, where observations are viewed as realizations from a population.
    The finite population perspective can be harder to grasp, particularly as $N \to \infty$.
    However, it can be helpful to think of the finite population framework as observing a sequence of numbers $\{Y_i(a) \mid i = 1, 2, \ldots\}$, where each $Y_i(a)$ is fixed in advance.
    From this perspective, letting $N \to \infty$ corresponds to observing the entire infinite sequence.

- Randomness only comes from A_i's
  - the treatment vector $\boldsymbol{A}$ is from a random permutation of N_1 1's and N_0 0's

$$\mathrm{E}(A_i) = \frac{N_1}{N}\ , \quad \mathrm{Var}(A_i) = \frac{N_1 N_0}{N^2}\ , \quad \mathrm{Cov}(A_i, A_j) = -\frac{N_1 N_0}{N^2(N-1)}\ , \quad (i \neq j)$$

Neyman's Repeated Sampling Approach

- Focus on estimating the ATE:

$$\mu(1) = \frac{1}{N}\sum_{i=1}^{N} Y_i(1) \ , \qquad \mu(0) = \frac{1}{N}\sum_{i=1}^{N} Y_i(0) \ , \qquad \tau_{\mathsf{ATE}} = \mu(1) - \mu(0) = \frac{1}{N}\sum_{i=1}^{N}\{Y_i(1) - Y_i(0)\}$$

- Neyman's procedure (classic frequentist's view):
  - Find an unbiased estimator of the ATE τ
  - Obtain variance and interval estimates of the unbiased estimator (derive the estimator's distribution under repeated sampling based on the randomization distribution of the assignment vector $\boldsymbol{A}$)

Neyman's Estimator

- Neyman's estimator of the ATE:

$$\widehat{\tau}_{\mathsf{ATE}} = \widehat{\mu}(1) - \widehat{\mu}(0) = \frac{1}{N_1} \sum_{i=1}^{N} A_i Y_i - \frac{1}{N_0} \sum_{i=1}^{N} (1 - A_i) Y_i$$

- Unbiased: $\mathrm{E}(\widehat{\tau}) = \tau$

Neyman's Estimator

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- Unbiased: $\mathrm{E}(\widehat{\tau}) = \tau$

$$\begin{aligned}
\mathrm{E}(\widehat{\tau}_{\mathsf{ATE}}) &= \mathrm{E}\left\{\frac{1}{N_1}\sum_{i=1}^{N} A_i Y_i - \frac{1}{N_0}\sum_{i=1}^{N}(1-A_i)Y_i\right\} \\
&= \mathrm{E}\left\{\frac{1}{N_1}\sum_{i=1}^{N} A_i Y_i(1) - \frac{1}{N_0}\sum_{i=1}^{N}(1-A_i)Y_i(0)\right\} && \text{SUTVA} \\
&= \frac{1}{N_1}\sum_{i=1}^{N} \mathrm{E}(A_i) Y_i(1) - \frac{1}{N_0}\sum_{i=1}^{N}\mathrm{E}(1-A_i)Y_i(0) && \text{CRE, and } Y_i(a) \text{ are fixed numbers} \\
&= \frac{1}{N}\sum_{i=1}^{N} Y_i(1) - \frac{1}{N}\sum_{i=1}^{N} Y_i(0) && \mathrm{E}(A_i) = N_1/N \\
&= \tau_{\mathsf{ATE}}
\end{aligned}$$

  - What is the expectation taken over? Randomness of the A_i's

Neyman's Variance Estimator

- One can show that

$$\mathrm{Var}(\widehat{\tau}_{\mathsf{ATE}}) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}$$

where

$$S_1^2 = \frac{1}{N-1}\sum_{i=1}^{N}\{Y_i(1) - \mu(1)\}^2 \ , \quad S_0^2 = \frac{1}{N-1}\sum_{i=1}^{N}\{Y_i(0) - \mu(0)\}^2 \ ,$$

$$S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}[\{Y_i(1) - Y_i(0)\} - \{\mu(1) - \mu(0)\}]^2$$

- S_1^2 is the variance of potential outcome $Y(1)$, S_0^2 is the variance of potential outcome $Y(0)$, and S_τ^2 is the variance of individual effects

Neyman's Variance Estimator

- One can show that

$$\mathrm{Var}(\widehat{\tau}_{\mathsf{ATE}}) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}$$

- Because we never observe both $Y_i(1)$ and $Y_i(0)$, S_τ^2 is not estimable from data, usually one just ignores S_τ^2, and considers a conservative variance estimator:

$$\widehat{\mathrm{Var}}(\widehat{\tau}_{\mathsf{ATE}}) := \frac{\widehat{S}_1^2}{N_1} + \frac{\widehat{S}_0^2}{N_0},$$

where $\widehat{S}_a^2$ is the sample variance for treatment group $A = a$

$$\widehat{S}_1^2 = \frac{1}{N_1 - 1} \sum_{i=1}^{N} A_i \{Y_i - \widehat{\mu}(1)\}^2, \quad \widehat{S}_0^2 = \frac{1}{N_0 - 1} \sum_{i=1}^{N} (1 - A_i) \{Y_i - \widehat{\mu}(0)\}^2$$

- Generally conservative because

$$\mathrm{E}\{\widehat{\mathrm{Var}}(\widehat{\tau}_{\mathsf{ATE}})\} = \mathrm{E}\left(\frac{\widehat{S}_1^2}{N_1}\right) + \mathrm{E}\left(\frac{\widehat{S}_0^2}{N_1}\right) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} = \mathrm{Var}(\widehat{\tau}) + \frac{S_\tau^2}{N} \geqslant \mathrm{Var}(\widehat{\tau}_{\mathsf{ATE}})$$

- Unbiased when the causal effects are constant across individuals, i.e., $Y_i(1) - Y_i(0)$ are the same for all i
- One can improve the variance estimator; see Aronow et al. (2014).

Confidence interval

- If there is no variation in individual causal effects, then $1-\alpha$ confidence interval is given by

$$\mathsf{CI} := \widehat{\tau}_{\mathsf{ATE}} \pm z_{1-\alpha/2}\widehat{\mathrm{Var}}(\widehat{\tau}_{\mathsf{ATE}}), \tag{1}$$

  where for $\alpha = 0.05$, $z_{1-\alpha/2} = 1.96$

- Finite-sample central limit theorem (Li and Ding, 2017)

$$\frac{\widehat{\tau}_{\mathsf{ATE}} - \tau_{\mathsf{ATE}}}{\sqrt{\mathrm{Var}(\widehat{\tau}_{\mathsf{ATE}})}} \to N(0,1)$$

  which implies the CI in (1) is conservative (wider) in general

- Can be used to test the weak null hypothesis $H_0 : \tau_{\mathsf{ATE}} = 0$

- This confidence interval covers τ with probability at least as large as $1-\alpha$ when the sample size is large enough

Regression analysis of the CRE

- OLS of the outcome Y on the treatment A with an intercept

$$Y_i = \alpha + \beta A_i + \epsilon_i \quad \Rightarrow \quad (\widehat{\alpha}, \widehat{\beta}) = \arg\min_{(a,b)} \sum_{i=1}^{n} (Y_i - a - bA_i)^2$$

- One can show that $\widehat{\alpha} = \widehat{\mu}(0)$ and $\widehat{\beta} = \widehat{\tau}_{\mathsf{ATE}} = \widehat{\mu}(1) - \widehat{\mu}(0)$

- Note: Although the OLS estimator is algebraically identical to Neyman's estimator, the underlying frameworks are different:
  - OLS is typically analyzed under an infinite population / i.i.d. sampling framework.
  - Neyman's estimator is analyzed under a finite population / randomization-based framework.

- the variance estimator from OLS (assuming $\mathrm{Var}(\epsilon_i) = \sigma^2$)

$$\widehat{\mathrm{Var}}_{\mathsf{OLS}}(\widehat{\beta}) = \frac{\sum_{i=1}^{N} \{Y_i - (1 - A_i)\widehat{\mu}(0) - A_i\widehat{\mu}(1)\}^2}{N - 2} \left(\frac{1}{N_1} + \frac{1}{N_0} \right) \simeq \frac{N_1 \widehat{S}_1^2 + N_0 \widehat{S}_0^2}{N} \frac{N}{N_1 N_0} = \frac{\widehat{S}_1^2}{N_0} + \frac{\widehat{S}_0^2}{N_1}$$

- the Eicker-Huber-White robust variance estimator (assuming $\mathrm{Var}(\epsilon_i) = \sigma_i^2$)

$$\widehat{\mathrm{Var}}_{\mathsf{EHW}}(\widehat{\beta}) \approx \frac{\widehat{S}_1^2}{n_1} + \frac{\widehat{S}_0^2}{n_0}$$

Proof: $\widehat{\beta} = \widehat{\tau}_{\mathsf{ATE}}$

$$
\begin{aligned}
\widehat{\beta} &= \frac{\sum_{i=1}^{N}(Y_i - \overline{Y})(A_i - \overline{A})}{\sum_{i=1}^{N}(A_i - \overline{A})^2} \\
&= \cdots \\
&= \frac{N}{N_1 N_0} \sum_{i=1}^{N} A_i Y_i - \frac{1}{N_0} \sum_{i=1}^{N} Y_i \\
&= \frac{N}{N_1} \sum_{i=1}^{N} A_i Y_i - \frac{1}{N_0} \sum_{i=1}^{N} (1 - A_i) Y_i \\
&= \widehat{\tau}_{\mathsf{ATE}}
\end{aligned}
$$

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Fisher's Null Hypothesis

- Fisher's sharp null hypothesis

$$H_0 : Y_i(1) = Y_i(0) \quad i = 1, \ldots, N$$

- No individual treatment effect



Binary Y: Fisher's Exact Test

Analysis of 2×2 table:

	$Y=1$	$Y=0$	
$A=1$	a	b	N_1
$A=0$	c	d	N_0
	$a+c$	$b+d$	N

- Row total is fixed due to the design: completely randomized experiment
- Column total is fixed under the sharp null: why?
  - $Y_i(1) = Y_i(0)$ under the sharp null;
    $Y_i = Y_i(1) = Y_i(0)$ regardless of A_i;
    the total number of units with $Y_i = Y_i(1) = Y_i(0) = 1$ is fixed in the finite-population framework
- a has a hypergeometric distribution if A and Y are independent (Lydersen et al., 2009)

Binary Y: Fisher's Exact Test

Analysis of 2×2 table:

	$Y = 1$	$Y = 0$	
$A = 1$	a	b	N_1
$A = 0$	c	d	N_0
	$a + c$	$b + d$	N

- Row sums and column sums are all fixed: once we know a, all other numbers are known

- Assume that A and Y are independent with fixed N_1, N_0:

  $\Pr(Y = 1 \mid A = 1) = \Pr(Y = 1 \mid A = 0) = \Pr(Y = 1) = p$

$$\Pr\{\#(A = 1, Y = 1) = a \mid \#(Y = 1) = a + c\} = \frac{\binom{a+b}{a} p^a (1-p)^b \binom{c+d}{c} p^c (1-p)^d}{\binom{n}{a+c} p^{a+c} (1-p)^{b+d}} = \frac{\binom{a+b}{a}\binom{c+d}{c}}{\binom{n}{a+c}}$$

- Example: Lady tasting tea

Example: Lady tasting tea

- The experiment was designed by Fisher in 1935 to test whether a lady could differentiate between two ways of making milk tea:
  - [Milk → Tea] OR [Tea → Milk]
- The lady claims that she can tell the difference
- Objective: determine if her performance is better than random guessing

Experiment setup

- **8 cups of tea** were prepared:
  - 4 cups with [Milk → Tea] and 4 cups with [Tea → Milk]
- The lady was asked to identify which 4 cups had milk added first
- Results were summarized in the following 2×2 table:

	Milk First (Lady)	Tea First (Lady)	Column Sum
Milk First (Fisher)	X	$4 - X$	4
Tea First (Fisher)	$4 - X$	X	4
Row Sum	4	4	8

- X: the number of cups correctly identified as "milk-first"
- X can take values: $0, 1, 2, 3, 4$.

Fisher's Exact Test

- Total number of possible combinations:
  Fisher chooses 4 "milk-first" cups out of 8 total cups

$$\binom{8}{4}$$

- Number of favorable combinations:
  The lady correctly identifies X "milk-first" cups out of the 4 actual "milk-first" cups, and $4 - X$ "tea-first" cups out of the 4 actual "tea-first" cups

$$\binom{4}{X}\binom{4}{4-X}$$

- Under the null hypothesis, X is based on the random guess, so

$$\Pr\left(X = x \,\middle|\, H_0\right) = \frac{\binom{4}{x}\binom{4}{4-x}}{\binom{8}{4}}$$

Fisher's Exact Test

- p-value = the probability of obtaining a test statistic at least as extreme as the one actually observed under H_0

$$p_{\text{FRT}} = \Pr\left(X \geqslant x \,\middle|\, H_0\right) = \sum_{t \geqslant x} \frac{\binom{4}{t}\binom{4}{4-t}}{\binom{8}{4}}$$

- In the actual experiment, the lady identified all 4 cups correctly $(X = 4)$.
  - This is the extremest case that can happen
  - p-value:

$$p_{\text{FRT}} = \frac{\binom{4}{4}\binom{4}{0}}{\binom{8}{4}} = \frac{1}{70} = 0.014$$

- If $X = 3$, then the p-value would be

$$p_{\text{FRT}} = \frac{\binom{4}{3}\binom{4}{1}}{\binom{8}{4}} + \frac{\binom{4}{4}\binom{4}{0}}{\binom{8}{4}} = \frac{16}{70} + \frac{1}{70} = 0.243$$

Fisher's Exact Test

- Why exact?
- the significance of the deviation from a null hypothesis (e.g., p-value) can be calculated exactly, rather than relying on an approximation that becomes exact in the limit as the sample size grows to infinity
- Fisher's exact test is a special case of Fisher Randomization Test, when both A and Y are binary
- In causal inference applications, A is typically binary, while Y may be continuous, categorical, or binary

Fisher Randomization Test

- Under sharp null hypothesis $H_0 : Y_i(1) = Y_i(0)\ , \quad i = 1, \ldots, N$,

$$\boldsymbol{Y}(1) = \boldsymbol{Y}(0) = (Y_1, \ldots, Y_N) = \boldsymbol{Y}$$

- Test statistic $T = T(\boldsymbol{A}, \boldsymbol{Y})$: $\boldsymbol{Y}$ is fixed under H_0, randomness only comes from $\boldsymbol{A}$
- What is the distribution of T?
  - recall that $\boldsymbol{A}$ is uniform over the set $\{\boldsymbol{a}^1, \ldots, \boldsymbol{a}^M\}$ where $M = \binom{N}{N_1}$
  - so T is uniform over the set $\{T(\boldsymbol{a}^1, \boldsymbol{Y}), \ldots, T(\boldsymbol{a}^M, \boldsymbol{Y})\}$
  - this distribution of T is called the *randomization distribution*

Fisher Randomization Test: Difference-in-means

- Test statistic: $D = \widehat{\tau}_{\mathsf{ATE}} = \frac{\sum_{i=1}^{N} A_i Y_i(1)}{\sum_{i=1}^{N} A_i} - \frac{\sum_{i=1}^{N} (1 - A_i) Y_i(0)}{\sum_{i=1}^{N} (1 - A_i)}$
- D is a random variable because of the randomness in A_i
- Want to get the distribution of D: get all possible randomization assignment (since they are equal likely)

Fisher Randomization Test: Example

- A new drug is being tested in humans for the first time to assess effect on CD4+ T cells in patients with HIV
- 7 individuals are randomized to 2 groups: control ($N - N_1 = 3$) or drug ($N_1 = 4$)
- Outcome $Y_i(a)$: percent increase in CD4+ count from baseline for $a = 0$ control, $a = 1$ drug
- Null hypothesis is the drug has no effect

$$H_0 : Y_i(1) = Y_i(0) \quad i = 1, ..., 7$$

Fisher Randomization Test: Example

- Observed data: control (65, 69, 73); drug (70, 88, 89, 92)
- There are $\binom{7}{4} = 35$ possible randomizations, i.e. there are 35 possible ways to assign 4 individuals to drug
- Under H_0, can compute D for each of these randomization assignment

Example: All Possible Group Assignments

$X=0$	$X=1$	d	$X=0$	$X=1$	d
65 69 70	73 88 89 92	-17.50	65 69 73	70 88 89 92	-15.75
65 69 88	70 73 89 92	-7.00	65 69 89	70 73 88 92	-6.42
65 69 92	70 73 88 89	-4.67	65 70 73	69 88 89 92	-15.17
65 70 88	69 73 89 92	-6.42	65 70 89	69 73 88 92	-5.83
65 70 92	69 73 88 89	-4.08	65 73 88	69 70 89 92	-4.67
65 73 89	69 70 88 92	-4.08	65 73 92	69 70 88 89	-2.33
65 88 89	69 70 73 92	4.67	65 88 92	69 70 73 89	6.42
65 89 92	69 70 73 92	6.00	69 70 73	65 88 89 92	-12.83
69 70 88	65 73 89 92	-4.08	69 70 89	65 73 88 92	-3.50
69 70 92	65 73 88 89	-1.75	69 73 88	65 70 89 92	-2.33
69 73 89	65 70 88 92	-1.75	69 73 92	65 70 88 89	0.00
69 88 89	65 70 73 92	7.00	69 88 92	65 70 73 89	8.75
69 89 92	65 70 73 88	9.33	70 73 88	65 69 89 92	-1.75
70 73 89	65 69 88 92	-1.17	70 73 92	65 69 88 89	0.58
70 88 89	65 69 73 92	7.58	70 88 92	65 69 73 89	9.33
70 89 92	65 69 73 88	9.92	73 88 89	65 69 70 92	9.33
73 88 92	65 69 70 89	11.08	73 89 92	65 69 70 92	10.67
88 89 92	65 69 70 73	20.42			

Fisher Randomization Test: Example

- Observed $D = 15.75$
- Two-sided p-value

$$P(|D| \geqslant 15.75) = 3/35 = 0.086$$

- In R

```
library(exactRankTests)
perm.test(c(65,69,73),c(70,88,89,92),exact=T)

library(coin)
yy <- c(65,69,73,70,88,89,92); xx <- c(0,0,0,1,1,1,1)
oneway_test(yy ~ as.factor(xx), distribution="exact")
```

Fisher Randomization Test: 3 steps

- Step 1) Specify the null hypothesis (sharp null)
- Step 2) Choose the statistic (e.g., difference-in-means)
- Step 3) Measure the extremeness by the p-value (under the randomization distribution)
  - one-sided (larger/smaller values are more extreme)
  - two-sided (distributed around zero)

Monte Carlo methods

- In practice, $M = \binom{N}{N_1}$ can be very large: $M > 10^{29}$ when $N = 100$ and $N_1 = 50$
- Computationally infeasible to enumerate all possible values of the treatment vector
- Monte Carlo methods
  - take B independent random draws from $\{\boldsymbol{a}^1, \ldots, \boldsymbol{a}^M\}$
  - in other words, randomly permute $\boldsymbol{A}$ for B times
  - approximate p_{FRT} by

$$\widehat{p}_{\mathsf{FRT}} = \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}\{T(\boldsymbol{a}^b, Y) \geqslant T(\boldsymbol{A}, Y)\}$$

$$\widehat{p}_{\mathsf{FRT}} = \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}\{|T(\boldsymbol{a}^b, Y)| \geqslant T(\boldsymbol{A}, Y)\}$$

- Also called the *permutation test*

Testing Other Sharp Nulls

- One can also test $H_0 : Y_i(1) - Y_i(0) = \tau_0 \ , \quad i = 1, \ldots, N$; under H_0:

$$\boldsymbol{Y}(1) = \boldsymbol{Y}\boldsymbol{A} + (\boldsymbol{Y} + \tau_0)(1 - \boldsymbol{A}) \ , \qquad \boldsymbol{Y}(0) = \boldsymbol{Y}(1 - \boldsymbol{A}) + (\boldsymbol{Y} - \tau_0)\boldsymbol{A}$$

- $H_0 : Y_i(1) - Y_i(0) = 5$:

i	$Y_i(1)$	$Y_i(0)$	A	Y
1	8	3=8-5	1	8
2	-1=-6-(-5)	-6	0	-6

- The test statistic $T = T(\boldsymbol{a}^m, \boldsymbol{Y}, \tau_0)$ becomes

$$\frac{1}{N_1}\sum_{i=1}^{N} a_i^m \boldsymbol{Y}_i(1) - \frac{1}{N_0}\sum_{i=1}^{N}(1 - a_i^m)\boldsymbol{Y}_i(0) = \frac{1}{N_1}\sum_{i=1}^{N} a_i^m \boldsymbol{Y}_i(1) - \frac{1}{N_0}\sum_{i=1}^{N}(1 - a_i^m)\boldsymbol{Y}_i(1) - \tau_0$$

- Again, the distribution test statistic $T = T(\boldsymbol{A}, \boldsymbol{Y}, \tau_0)$ can be obtained under H_0 by leveraging $\boldsymbol{A} \sim$ CRE

Confidence Intervals from Testing Sharp Nulls

- From statistics courses, there is a duality between testing a hypothesis and $H_0 : Y_i(1) - Y_i(0) = \tau_0$ and constructing confidence intervals of the parameter $Y_i(1) - Y_i(0) = \tau$
  - If the null value τ_0 is inside a $1 - \alpha$ confidence interval, you retain H_0; otherwise, you reject H_0
  - If you consider a set of τ_0s where its corresponding $H_0$and associated p-value is above α (i.e. the set of H_0s that were retained), this set of τ_0s is the $1 - \alpha$ confidence interval.

- You can use the second fact to construct a $1 - \alpha$ confidence interval for the parameter $Y_i(1) - Y_i(0) = \tau$. Specifically, we can iterate the above procedure for different sharp nulls and retain τ_0s whose associated p-values is above α

- Of course, the above assumes that every unit i has the same causal effect $Y_i(1) - Y_i(0) = \tau$ (i.e., homogeneity)

- Therefore, if all of our sharp null hypotheses are consistently rejected (i.e., the p-value is below α for all $\tau \in \mathbb{R}$), the observed data do not support the sharp null scenario and suggest that individuals have heterogeneous causal effects. Consequently, the confidence interval obtained from the above procedure would be empty.

Wilcoxon Rank Sum

- Useful when some Y_i values are outliers
- Based on the ranks of the pooled observed outcomes
- Denote the rank of Y_i by R_i: $R_i = \#\{j : Y_j \leqslant Y_i\}$
- Statistic: $W = \sum_{i=1}^{n} A_i R_i$
- Since the sum of the ranks in the pooled samples is fixed: $1 + 2 + \cdots + n = n(n+1)/2$, W is equivalent to the difference in the means of the ranks under treatment/control
- Also related to Hodges and Lehmann (1963)
- `R` function `wilcox.test`: can compute both exact and asymptotic p-values

Neymanian vs. Fisherian Inference

- Neymanian inference:
  - weak null hypothesis: $\tau = 0$
  - only about the difference in mean

- Fisherian inference:
  - sharp null hypothesis: $Y_i(1) = Y_i(0)$ for $i = 1, \ldots, N$
  - works for any test statistic

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Covariates in Randomized Experiments

$$(A_i, Y_i, X_i) \quad i = 1, \ldots, N \qquad \text{where } X_i \text{ are covariates}$$

- Randomization aims to balance all baseline covariates (both observed and unobserved) on average
- Assess whether the randomization was successful by examining the distributions of $X|A = 1$ and $X|A = 0$
  - Draw histogram/boxplot; Conduct 2-sample t-test
- Duflo et al. (2015)

TABLE 1—BASELINE CHARACTERISTICS, BY TREATMENT GROUP

	Stand-alone education subsidy (S)	Stand-alone HIV education (H)	Joint program (SH)	Control (C)	Balance tests: p-value for test that: $S = C$	$H = C$	$SH = C$	$SH = S$ $SH = H$
Panel A. Baseline characteristics of schools								
Average score on primary school graduation exam in 2003	255.2 [29.6]	249.4 [24.8]	248.6 [32.04]	249.3 [26.1]	*0.097**	*0.66*	*0.468*	*0.146* *0.857*
School size (total number of students)	464.6 [203.1]	489.3 [208.8]	473.8 [185.7]	498.9 [194.3]	*0.292*	*0.777*	*0.587*	*0.764* *0.611*
Sex ratio (female/male) among students in 2002	1.016 [0.124]	1.024 [0.127]	1.012 [0.105]	1.016 [0.135]	*0.945*	*0.455*	*0.823*	*0.857* *0.556*
Number of latrines on school compound	11.6 [6.3]	11.2 [6.4]	9.9 [5.7]	11.1 [5.5]	*0.215*	*0.635*	*0.081**	*0.068** *0.173*
Number of primary schools within 2 kilometer radius	2.01 [1.95]	2.16 [1.82]	2.06 [1.8]	2.06 [1.76]	*0.845*	*0.521*	*0.925*	*0.86* *0.742*
Total number of teachers in 2003	14.2 [4.2]	14.6 [5.3]	13.8 [4.4]	14.6 [4.7]	*0.786*	*0.439*	*0.282*	*0.613* *0.257*
Average age of teachers in 2003	40.0 [3.1]	39.6 [3.8]	39.6 [3.8]	39.6 [3.5]	*0.455*	*0.796*	*0.845*	*0.57* *0.975*
Sex ratio (female/male) among teachers in 2003	1.22 [1.0]	1.18 [0.848]	1.30 [0.987]	1.15 [0.829]	*0.856*	*0.788*	*0.281*	*0.575* *0.405*

Covariates in Randomized Experiments

$$(A_i, Y_i, X_i) \quad i = 1, \ldots, N \qquad \text{where } X_i \text{ are covariates}$$

- However, covariate imbalance can still occur due to random chance
- Why is covariate balance important?
- Better balance can
  - provide more meaningful estimates of causal effects
  - increase the precision of estimated causal effects

Covariates in Randomized Experiments

- Option 1: force better balance on important covariates by design
  - stratified randomized experiments
  - rerandomization

- Option 2: correct imbalance in covariates by analysis
  - post-stratification
  - regression adjustment (ANCOVA)

	design	analysis
discrete covariate	stratification	post-stratification
general covariate	rerandomization	regression adjustment

Stratified Randomized Experiments

- Divide subjects into K strata based on covariate $X_i \in \{1, \ldots, K\}$
- Conduct completely randomized trials within each stratum

Define for $k = 1, \ldots, K$

$$N_{[k]1} = \#\{i : X_i = k, A_i = 1\}, \quad N_{[k]0} = \#\{i : X_i = k, A_i = 0\}$$

$$N_{[k]} = \#\{i : X_i = k\} = N_{[k]1} + N_{[k]0}, \quad \pi_{[k]} = N_{[k]}/N$$

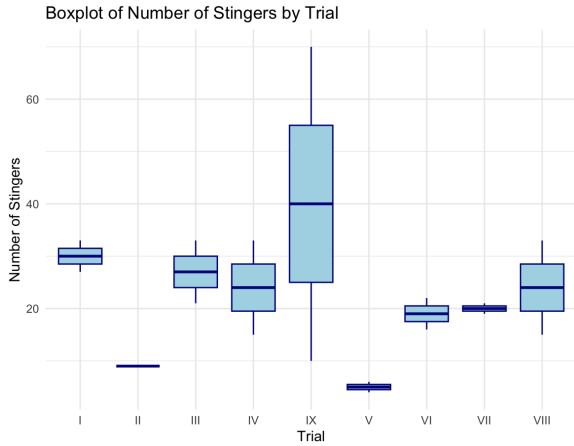
- Total number of randomizations in a SRE is $\prod_{k=1}^{K} \binom{N_{[k]}}{N_{[k]1}}$
- Stratum-specific average causal effect: $\tau_{[k]} = \sum_{X_i = k} \{Y_i(1) - Y_i(0)\}/N_{[k]}$
- Average causal effect: $\tau = \sum_{k=1}^{K} \pi_{[k]} \tau_{[k]}$

SRE: Example

- Free (1961)
- If you are stung by a bee, does that make you more likely to get stung again? (Bees might leave behind a chemical message that tells other bees to attack you)
- Scientists dangled 16 muslin-wrapped cotton balls over a beehive, where half of the balls had been previously stung and the other half were fresh
- Outcome: total number of new stings
- Repeated for a total of nine trials

SRE: example

- Scientists expect the number of stings to vary by trial
  (different number of bees in the hive, different times of day, different weather, etc.)
- Each trial is a different strata ($K = 9$)

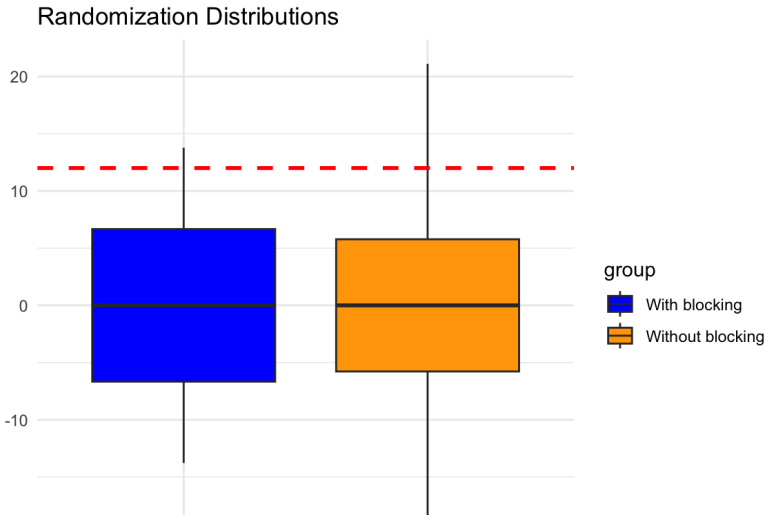


SRE: Fisherian inference

- Sharp null hypothesis: $H_0 : Y_i(1) = Y_i(0)$ for $i = 1, \ldots, N$
- Stratified estimator: $\widehat{\tau}_S = \sum_{k=1}^{K} \pi_{[k]} \widehat{\tau}_{[k]}$ where
  $\widehat{\tau}_{[k]} = N_{[k]1}^{-1} \sum_{i=1}^{N} \mathbb{1}(X_i = k, A_i = 1) Y_i - N_{[k]0}^{-1} \sum_{i=1}^{N} \mathbb{1}(X_i = k, A_i = 0) Y_i$ is the difference-in-means in stratum k
- Simulate randomization schemes compatible with the randomization design, in this case SRE
- When N is large, it is easier to simulate randomizations rather than enumerate all possible allocations

SRE: Fisherian inference

- $\widehat{\tau}_S = 12$
- Two-sided p-value: 0.078 (with blocking), 0.087 (without blocking)



SRE: Neymanian inference

- null hypothesis: $\tau = N^{-1}\sum_{i=1}^{N}\{Y_i(1) - Y_i(0)\} = 0$
- within stratum k

$$\widehat{\tau}_{[k]} = \frac{1}{N_{[k]1}}\sum_{i=1}^{N}\mathbb{1}(X_i = k, A_i = 1)Y_i - \frac{1}{N_{[k]0}}\sum_{i=1}^{N}\mathbb{1}(X_i = k, A_i = 0)Y_i$$

$$\mathrm{Var}(\widehat{\tau}_{[k]}) = \frac{S^2_{[k]1}}{N_{[k]1}} + \frac{S^2_{[k]0}}{N_{[k]0}} - \frac{S^2_{[k]\tau}}{N_{[k]}}$$

- stratified estimator:

$$\widehat{\tau}_S = \sum_{k=1}^{K}\pi_{[k]}\widehat{\tau}_{[k]}, \quad \mathrm{Var}(\widehat{\tau}_S) = \sum_{k=1}^{K}\pi^2_{[k]}\mathrm{Var}(\widehat{\tau}_{[k]})$$

- a conservative variance estimator:

$$\widehat{\mathrm{Var}}(\widehat{\tau}_S) = \sum_{k=1}^{K}\pi^2_{[k]}\left(\frac{\widehat{S}^2_{[k]1}}{N_{[k]1}} + \frac{\widehat{S}^2_{[k]0}}{N_{[k]0}}\right)$$

Why SRE?

- *"Block what you can; randomize what you cannot"* – George Box
- efficiency gain when the stratification variable is predictive of potential outcomes (compared to CRE)
- no harm when the stratification variable is not predictive
- math proof in Ding's textbook §5.3.3 (Ding, 2024)
- based on large stratum assumption (cannot arbitrarily increase K)
- difficult to extend to many covariates or continuous covariates

Covariates in Randomized Experiments

- Option 1: force better balance on important covariates by design
  - stratified randomized experiments
  - rerandomization

- Option 2: correct imbalance in covariates by analysis
  - post-stratification
  - regression adjustment (ANCOVA)

	design	analysis
discrete covariate	stratification	post-stratification
general covariate	rerandomization	regression adjustment

Post-stratification

- Math: the conditional distribution of $\boldsymbol{A}$ from a CRE given $\boldsymbol{N} = \{N_{[k]1}, N_{[k]0}\}_{k=1}^{K}$ is identical to the distribution of $\boldsymbol{A}$ from an SRE
- Idea: conditional on $\boldsymbol{N}$, we can analyze a CRE with a discrete covariate X in the same way as in a SRE
- Note: $\boldsymbol{N}$ is a fixed quantity in a SRE, but $\boldsymbol{N}$ is a random quantity in a CRE; but the post-stratification is motivated to view $\boldsymbol{N}$ as a fixed quantity (=variable in the conditioning argument)
- Fisher Randomization Test: simulate treatment vector compatible with $\{N_{[k]1}, N_{[k]0}\}_{k=1}^{K}$
- Neymanian inference: $\widehat{\tau}_{PS} = \sum_{k=1}^{K} \pi_{[k]} \widehat{\tau}_{[k]} = \widehat{\tau}_S$, variance of $\widehat{\tau}_{PS}$ conditional on $\mathbf{n}$ is the same as $\mathrm{Var}(\widehat{\tau}_S)$
- Stratification uses X in the design stage and post-stratification uses X in the analysis stage. Asymptotically, their difference is small with large strata

Covariates in Randomized Experiments

- Option 1: force better balance on important covariates by design
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  - rerandomization

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  - regression adjustment (ANCOVA)

	design	analysis
discrete covariate	stratification	post-stratification
general covariate	rerandomization	regression adjustment

Rerandomization

- Conceptually, a stratified randomized experiment can be viewed as a two-step procedure:
  - Step 1: Generate $\boldsymbol{A}$ using a completely randomized design
  - Step 2: Check covariate balance within each stratum k. If there exists a stratum k s.t.

$$\sum_{i=1}^{N} \mathbb{1}\{X_i = k, A_i = 1\} \neq \sum_{i=1}^{N} \mathbb{1}\{X_i = k, A_i = 0\}$$

    reject $\boldsymbol{A}$ and return to Step 1

- Rerandomization is analogous, except that the discrepancy metric is adapted to accommodate continuous covariates
  - Step 1: Generate $\boldsymbol{A}$ using a completely randomized design
  - Step 2: Check covariate balance within each stratum k. k. If $M(\boldsymbol{A}, \boldsymbol{X}) > t$ for some constant t, reject $\boldsymbol{A}$ and return to Step 1

- Morgan and Rubin (2012): Mahalanobis distance $M(\boldsymbol{A}, \boldsymbol{X}) = (\overline{\boldsymbol{X}}_1 - \overline{\boldsymbol{X}}_0)^\intercal \mathrm{Cov}(\boldsymbol{X})^{-1} (\overline{\boldsymbol{X}}_1 - \overline{\boldsymbol{X}}_0)$

- Rerandomization avoids covariate imbalance by discarding the treatment allocations with large difference between covariates in the treatment/control group

Covariates in Randomized Experiments

- Option 1: force better balance on important covariates by design
  - stratified randomized experiments
  - rerandomization

- Option 2: correct imbalance in covariates by analysis
  - post-stratification
  - regression adjustment (ANCOVA)

	design	analysis
discrete covariate	stratification	post-stratification
general covariate	rerandomization	regression adjustment

Covariate-adjusted Fisher Randomization Test

- Sharp null $H_0 : Y_i(1) = Y_i(0)$ for $i = 1, \ldots, N$
- OLS (or other model) fitting of Y_i on X_i
- Obtain residual ϵ_i
  - Y_i and X_i are all fixed; therefore, values of ϵ_i are fixed
- Replace Y_i by ϵ_i in the test statistic $T = T(A, Y, X)$
  - or replace R_i by the rank of ϵ_i for Wilcoxon rank sum test
- Note that OLS can be replaced by other models or even machine learning algorithms
  - However, with black-box models, it may be unclear whether the ϵ_i values can be regarded as fixed

Covariate Adjustment: ANCOVA

- Null hypothesis $H_0 : \tau = N^{-1} \sum_{i=1}^{N} \{Y_i(1) - Y_i(0)\} = 0$
- Run OLS regression of Y_i on A_i, X_i, and possibly $A_i \cdot X_i$
- Use the coefficient of A_i as the adjusted estimator of ATE τ
- Also known as the ANCOVA estimator
- Two different ANCOVA models:
  - ANCOVA 1 (Fisher (1935) without interactions): $Y_i \sim \beta_0 + \beta_a A_i + \beta_x X_i$
  - ANCOVA 2 (Lin (2013) with interactions): $Y_i \sim \beta_0 + \beta_a A_i + \beta_x X_i + \beta_{ax} A_i X_i$
- In both models, the OLS estimator of β_a is taken as the estimator of τ, denoted as $\widehat{\tau}_{\mathsf{F}}$ (ANCOVA 1) and $\widehat{\tau}_{\mathsf{L}}$ (ANCOVA 2)
- Question: Any difference between the two models? Which one is better?

ANCOVA Adjustment: Key Points

- Lin (2013)

- To obtain an ANCOVA estimator that is more efficient than the unadjusted estimator of ATE τ in randomized experiments, one must:

1. Include a full set of treatment-covariates interaction (use ANCOVA 2)
2. Center covariates (so you can directly use the OLS estimate of the coefficient of A in ANCOVA as the adjusted estimator for τ)

Key Idea: Reducing Variance of ATE Estimator

- Neyman (1923) showed that the variance of the difference-in-means estimator $\widehat{\tau}$ depends on the variances of potential outcomes
- Intuitively, we can reduce the variance by reducing the variances of the potential outcomes
- A family of covariate-adjusted estimators:

$$\begin{aligned}\widehat{\tau}(\beta_1, \beta_0) &= \frac{1}{N_1}\sum_{i=1}^{N} A_i(Y_i - \beta_1^\intercal X_i) - \frac{1}{N_0}\sum_{i=1}^{N}(1 - A_i)(Y_i - \beta_0^\intercal X_i) \\ &= \{\widehat{\mu}_Y(1) - \beta_1^\intercal \widehat{\mu}_X(1)\} - \{\widehat{\mu}_Y(0) - \beta_0^\intercal \widehat{\mu}_X(0)\}\end{aligned}$$

Interpretation of the Adjusted Estimator

- We transform the outcomes to:

$$Y_i(1) - \beta_1^\intercal X_i, \quad Y_i(0) - \beta_0^\intercal X_i$$

- The new estimator is the difference in means of these adjusted outcomes
- When $\beta_1 = \beta_0 = 0$, we recover the simple difference-in-means estimator $\widehat{\tau}$
- Assume that $\overline{X} = 0$, then for any fixed β_1 and β_0,

$$\mathrm{E}\{\widehat{\tau}(\beta_1, \beta_0)\} = \sum_{i=1}^{N} \{Y_i(1) - Y_i(0)\} = \tau$$

- need to center X, otherwise $\mathrm{E}\{\widehat{\tau}(\beta_1, \beta_0)\} = \tau - (\beta_1^\intercal - \beta_0^\intercal)\overline{X} \neq \tau$

Variance Formula for the Adjusted Estimator

$$\mathrm{Var}(\hat{\tau}(\beta_1, \beta_0)) = \frac{S_1^2(\beta_1)}{N_1} + \frac{S_0^2(\beta_0)}{N_0} - \frac{S_\tau^2(\beta_1, \beta_0)}{N}$$

- $S_a^2(\beta_a)$ and $S_\tau^2(\beta_1, \beta_0)$ = finite population variances of adjusted potential outcomes and individual causal effects

- a conservative variance estimator is

$$\widehat{\mathrm{Var}}(\hat{\tau}(\beta_1, \beta_0)) = \frac{\widehat{S}_1^2(\beta_1)}{N_1} + \frac{\widehat{S}_0^2(\beta_0)}{N_0}$$

where

$$\widehat{S}_1^2(\beta_1) = \frac{1}{N_1 - 1} \sum_{i=1}^{N} A_i (Y_i - \alpha_1 - \beta_1^\intercal X_i)^2$$

$$\widehat{S}_0^2(\beta_0) = \frac{1}{N_0 - 1} \sum_{i=1}^{N} (1 - A_i)(Y_i - \alpha_0 - \beta_0^\intercal X_i)^2$$

- α_1 and α_0 are the sample means of $Y_i - \beta_1^\intercal X_i$ under treatment and $Y_i - \beta_0^\intercal X_i$ under control

OLS Estimation of (β_1, β_0)

- To minimize $\widehat{\mathrm{Var}}(\hat{\tau}(\beta_1, \beta_0))$, we solve two OLS problems:

$$\min_{\alpha_1, \beta_1} \sum_{i=1}^{n} A_i (Y_i - \alpha_1 - \beta_1^\intercal X_i)^2$$

$$\min_{\alpha_0, \beta_0} \sum_{i=1}^{n} (1 - A_i)(Y_i - \alpha_0 - \beta_0^\intercal X_i)^2$$

- Run OLS separately for treatment and control groups

- Estimate (α_1, β_1) for the treatment group and (α_0, β_0) for the control group

Final Estimator: Lin (2013)

- **Final Estimator:**

$$\begin{aligned}\widehat{\tau}(\widehat{\beta}_1, \widehat{\beta}_0) &= \frac{1}{N_1}\sum_{i=1}^{N} A_i(Y_i - \widehat{\beta}_1^\intercal X_i) - \frac{1}{N_0}\sum_{i=1}^{N}(1-A_i)(Y_i - \widehat{\beta}_0^\intercal X_i) \\ &= \left\{\widehat{\mu}_Y(1) - \widehat{\beta}_1^\intercal \widehat{\mu}_X(1)\right\} - \left\{\widehat{\mu}_Y(0) - \widehat{\beta}_0^\intercal \widehat{\mu}_X(0)\right\} \\ &= \widehat{\alpha}_1 - \widehat{\alpha}_0\end{aligned}$$

- This is equivalent to the coefficient of A_i in the OLS regression of Y_i on $(1, A_i, X_i, A_i X_i)$, which is $\widehat{\tau}_L$

  Asymptotically, $\widehat{\tau}_{\text{L}}$ (ANCOVA 2) is more efficient than $\widehat{\tau}$ (the unadjusted estimator)

Why Do Interactions Matter?

- For simplicity, consider a single centered continuous covariate X with $\bar{X} = 0$
- Build prediction models for $Y(1)$ and $Y(0)$ based on X using data from the treatment/control group:

$$\widehat{\mu}_1(X) = \widehat{\alpha}_1 + \widehat{\beta}_1 X \quad \widehat{\mu}_0(X) = \widehat{\alpha}_0 + \widehat{\beta}_0 X$$

- Then one possible estimator for τ is

$$\frac{1}{N}\sum_{i=1}^{N}\{\widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i)\} = \widehat{\alpha}_1 - \widehat{\alpha}_0$$

- Use the equation $Y_i = Y_i(1)A_i + Y_i(0)(1 - A_i)$, a prediction model for Y based on X would be

$$\begin{aligned}\widehat{\mu}(X) &= (\widehat{\alpha}_1 + \widehat{\beta}_1 X)A + (\widehat{\alpha}_0 + \widehat{\beta}_0 X)(1 - A) \\ &= \widehat{\alpha}_0 + (\widehat{\alpha}_1 - \widehat{\alpha}_0)A + \widehat{\beta}_0 X + (\widehat{\beta}_1 - \widehat{\beta}_0)XA\end{aligned}$$

Model-free Validity for Model-based Statistics

- $\widehat{\tau}_{\mathrm{L}}$ is biased for τ in finite samples (the bias is usually small and negligible), but is consistent
- Model-free validity for model-based statistics:
  consistency of $\widehat{\tau}_{\mathrm{L}}$ holds even if the linear regression assumption is wrong
- Intuition
  - By randomization, the super-population correlation between A and X is zero
  - The correlation vanishes in large sample, but may not in finite sample
- Only holds in randomization - not true in observational studies!

Variance

- To estimate the variance of $\hat{\tau}_{\text{L}}$ under randomization, use the Eicker-Huber–White robust variance estimator (also known as the heteroscedasticity-consistent estimator)

$$\widehat{\text{Var}}(\hat{\tau}_{\text{L}}) = (W^\intercal W)^{-1}\{W^\intercal \text{diag}(\hat{u}_1^2, \cdots, \hat{u}_n^2)W\}(W^\intercal W)^{-1}$$

  where W is the matrix of regressors and $\hat{u}_i$ are the regression residuals

- $\widehat{\text{Var}}(\hat{\tau}_{\text{L}})$ is consistent or asymptotically conservative
  - When linear regression assumption holds: consistent
  - Otherwise, asymptotically conservative

- With two treatment groups of equal size, the conventional OLS variance estimator is also consistent or asymptotically conservative

OLS Adjustment: History

- Fisher (1935) proposed ANCOVA estimator in randomized experiments, argued that $\widehat{\tau}_{\text{F}}$ improves the precision (variance) over the unadjusted estimator $\widehat{\tau}$
- Freedman (2008a,b) provided an opposing view: gave examples where $\widehat{\tau}_{\text{F}}$ decrease precision (unbalanced experiments with heterogeneity, non-continuous outcome)
- Lin (2013) provided an agnostic view: re-examined Freedman's critique and proposed $\widehat{\tau}_{\text{L}}$ – case closed
- Tsiatis and co-authors made similar findings to Lin earlier (Yang and Tsiatis, 2001; Tsiatis et al., 2008)

Outline

1. Observational Study vs Randomized Experiment
2. Randomized Experiment
3. Examples of Randomized Experiments
4. Model-based Inference: Infinite Population
5. Randomization-based Inference: Finite Population: Neymanian Inference
6. Randomization-based Inference: Finite Population: Fisherian Inference
7. Covariate Balance and Adjustment
8. Assignment

Assignment

- Analyze the dataset `causalweight::JC` (Job Corps)
  While the dataset contains additional variables, we will focus only on the following:
  - $X =$ (`female, age, white, black, hispanic`)
  - $A =$ `assignment`
  - $Y =$ `earny4`

  We aim to study the effect of Job Corps program assignment on weekly earnings in the fourth year after assignment

- For detailed variable descriptions/contexts, see `?causalweight::JC`

- Using the dataset, check covariate balancing and conduct the analysis:
  - using model-based inference;
  - using Neymanian inference;
  - using Fisherian inference; (due to the large sample size, you may need to use a Monte Carlo approach; also construct confidence intervals)
  - using ANCOVA

- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

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